

Topic: Cartesian Product Operation

Q. What is a Cartesian Product in DBMS Explain its working mechanism and provide an example demonstrating how it combines two relations.

[DEF] Definition to Remember

The Cartesian Product (also called Cross Product, denoted $\times$) is a binary operation in relational algebra that pairs every row of the first relation with every row of the second relation, producing all possible combinations. Output columns = sum of both columns; Output rows = product of both row counts. Union Compatibility is NOT required.

Cartesian Product Operation ($\times$)

* Binary operation -- operates on two tables (which need NOT be union compatible).

2. Working of Cartesian Product

* Takes every single row from A and pairs it with every single row from B.

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A x B :
Row1_A <--> Row1_B
Row1_A <--> Row2_B
Row2_A <--> Row1_B
Row2_A <--> Row2_B
(2 x 2 = 4 combinations)
```

3. Example

COLORS (2 rows)

Color_ID	Color_Name
1	Red
2	Blue

SHAPES (2 rows)

Shape_ID	Shape_Name
A	Circle
B	Square

Query: COLORS x SHAPES

Color_ID	Color_Name	Shape_ID	Shape_Name
1	Red	A	Circle
1	Red	B	Square
2	Blue	A	Circle
2	Blue	B	Square

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Cartesian Product ($\times$) = Cross Product -- pairs every row of A with every row of B.
2. Output columns = Columns_A + Columns_B; Output rows = Rows_A x Rows_B.
3. Union Compatibility is NOT required -- the two relations can be completely different.
4. On its own, produces large amounts of meaningless data.
5. It is the foundation of JOIN operations: Cartesian Product + Selection = Join.
6. If A has m rows and B has n rows, the result has m x n rows.
7. Cartesian Product is a basic (fundamental) operator in relational algebra.

[RECALL] Quick Recall

$\times$ (Cartesian Product) -> Binary -> All Row Combinations -> Cols = A+B, Rows = AxB -> No Union Compatibility Needed -> Foundation of JOIN